03 · Case 3, part 1: Ane's asthma attack

The words of this track, as it says them. Bold brown is the examiner.

Exam case 3. Asthma.

Ane has had severe asthma since they were young. During a particularly bad attack, Ane has difficulty breathing, particularly on the out-breath. Ane's asthma is often triggered by small particles of pollen that make their way into the lungs, despite the protective features designed to keep them out. Ane makes sure to have their Salbutamol inhaler, which helps relieve the symptoms of an attack.

Case 3 is the biggest case in the exam, so it's split over three tracks. This one is the attack itself.

One mark. What term is used to describe difficulty breathing?

Dyspnoea. Dys means bad, and pnoea means breathing. Compare eupnoea, which is normal breathing. Tachypnoea, fast. Apnoea, none at all. And hyperventilation, breathing hard and fast, which changes the blood's carbon dioxide.

Three marks. What is happening in Ane's lungs to make breathing difficult?

Three things, a mark each. The hook: squeeze, swell, slime.

Bronchoconstriction: the smooth muscle around the airway has contracted, so the airway is constricted. Inflammation: the wall of the airway is thick with swelling. And increased mucus secretion, which leaves less room for air.

Two marks. How are Ane's ventilation and airway resistance affected?

Airway resistance is high, so it's harder to breathe. It takes a greater pressure difference to move the air, and that's the dyspnoea. And the accessory respiratory muscles are recruited: extra muscles for breathing. Her phrase for it is an increase in the work of breathing.

One mark. How does an inhaler help when Ane is having an asthma attack?

A reliever inhaler, like Salbutamol, is a bronchodilator. It forces the smooth muscle of the airways to relax, which lets the airway widen, and that relieves the symptoms so Ane can breathe. Don't mix it up with a preventer, which is a corticosteroid: an anti-inflammatory.

Her formative test asks the same things as drop-downs. Difficulty breathing: dyspnoea. The bronchi and bronchioles become obstructed by swelling and mucus accumulation. And the inhaler makes the airways dilate.

Once more. Ane's attack.

Difficulty breathing is dyspnoea: dys, bad, pnoea, breathing. Three things narrow Ane's airways: bronchoconstriction, with the smooth muscle contracted; inflammation, a swollen airway wall; and more mucus. Airway resistance is high, so it needs a greater pressure

difference, and the accessory muscles are recruited: more work of breathing. The reliever is a bronchodilator: it relaxes the airway smooth muscle so the airway widens. A preventer is a corticosteroid, an anti-inflammatory.